

```
diff --git a/README.md b/README.md
index cdbd965..284a4bf 100644
--- a/README.md
+++ b/README.md
@@ -3,7 +3,7 @@
```

> **LexPulse: Mitigating Amendment Blindness in E-Governance Legal QA using Adaptive Relationship-Aware**

-A research-grade AI platform that accurately answers questions about Indian legal policy documents – including legal amendments

+A research-grade AI platform that answers questions about Indian legal policy documents – including legal amendments

```
@@ -15,7 +15,7 @@ A research-grade AI platform that accurately answers questions about Indian legal policy documents – including legal amendments

| Naive RAG | 54.9% | 44.4% ■■ |
| Adaptive RAG (LexPulse) | 70.6% ■■■ | 72.2% ■■■ |
```

-> LexPulse achieves a **+15.7 percentage point** improvement over Naive RAG on Amendment-Trap queries, including legal amendments

+> LexPulse achieves a **+15.7 percentage point** overall accuracy improvement over Naive RAG, and a **+15.7 percentage point** improvement on Amendment-Trap queries, including legal amendments

```
@@ -23,13 +23,35 @@ A research-grade AI platform that accurately answers questions about Indian legal policy documents – including legal amendments
```

Standard RAG systems suffer from **"Amendment Blindness"** – they retrieve the original version of a document rather than the latest amendment.

-1. **Policy Amendment Knowledge Graph** (``data/relationship_graph.json``)

- A JSON graph where nodes = Acts, edges = ``amended_by`` relationships
- At query time, the retriever **traverses the graph** to automatically inject amendment documents into the search results

+1. **JSON-encoded Policy Amendment Relationship Graph** (``data/relationship_graph.json``)

- + A JSON graph where nodes = documents and edges encode amendment/supersession relationships
- + At query time, the retriever uses this graph metadata to inject active amendment documents that v

Implementation details – Relationship Graph (JSON metadata)

+The Policy Amendment Relationship Graph is encoded as JSON metadata rather than stored in a graph-database

+At query time, hybrid BM25 + vector retrieval produces candidate chunks. For any retrieved base document, we retrieve all amendments that v

+Pseudocode (one-hop injection):

```
+
+```python
+def inject_amendments(candidates, graph, vectorstore, k=3):
+    augmented = list(candidates)
+    candidate_sources = {c['source'] for c in candidates}
+    for c in candidates:
+        doc_id = c['source']
+        for amend_id in graph.get(doc_id, {}).get('amended_by', []):
+            if amend_id not in candidate_sources:
+                amendment_chunks = vectorstore.similarity_search_by_source(amend_id, top_k=k)
+                augmented.extend(amendment_chunks)
+                candidate_sources.update(ch['source'] for ch in amendment_chunks)
+    return augmented
+```
```

2. **Adaptive LLM Fallback**

- - When the vector database has no relevant context, instead of hallucinating, LexPulse gracefully
- - This eliminates the "RAG Penalty" – the accuracy drop that Naive RAG suffers on general questions
- + - When the vector database has no relevant context, LexPulse gracefully falls back to the LLM's int
- + - This helps reduce the typical RAG accuracy penalty found in some baseline deployments

@@ -41,14 +63,14 @@ LexPulse/

```
■■■ frontend/
■   ■■■ index.html           # Web dashboard UI
■■■ src/
-■   ■■■ retrieval_engine.py  # Adaptive RAG + Knowledge Graph injection
+■   ■■■ retrieval_engine.py  # Adaptive RAG + relationship graph injection
■   ■■■ evaluate.py          # LLM-as-a-Judge evaluation suite (51 Qs)
■   ■■■ ingestion.py         # PDF chunking + metadata enrichment
■   ■■■ metadata_tagger.py   # Amendment relationship tagger
■   ■■■ results_manager.py   # Eval results persistence
■   ■■■ auth_db.py           # User authentication
■■■ data/
-■   ■■■ relationship_graph.json # Policy Amendment Knowledge Graph
+■   ■■■ relationship_graph.json # Policy Amendment Relationship Graph
■   ■■■ eval_results_v2.json   # Latest evaluation results
```

```
■ ■■■ raw/ # Source PDF directory (PDFs not included)
■■■ generate_eval_chart.py # Publication-quality results chart generator
```

```
@@ -125,7 +147,7 @@ Submitted to: [Conference Name]
```

```
| Vector Store | ChromaDB |
| Embeddings | HuggingFace `all-MiniLM-L6-v2` |
| Hybrid Retrieval | BM25 + Semantic Search |
-| Knowledge Graph | Custom JSON graph (`relationship_graph.json`) |
+| Relationship Graph | Custom JSON graph (`relationship_graph.json`) |
| Backend | Python `http.server` |
| Frontend | Vanilla HTML/CSS/JS + TailwindCSS |
```

```
diff --git a/docs/index.html b/docs/index.html
```

```
index 1093855..8aa74a3 100644
```

```
--- a/docs/index.html
```

```
+++ b/docs/index.html
```

```
@@ -197,6 +197,21 @@
```

```

    <i data-lucide="database" class="w-16 h-16 mx-auto text-teal-500 mb-4"></i>
    <h3 class="text-xl font-bold text-slate-800 mb-2">Synchronize Raw Policy Data</h3>
    <p class="text-slate-500 font-medium mb-8">This will process the PDF repository, extract
+    <details class="text-left mx-auto max-w-2xl mt-4 bg-slate-50 p-4 rounded-lg border border-slate-200"
+        <summary class="font-semibold text-slate-700 cursor-pointer">Technical note: Relationship Graph</summary>
+        <p class="text-sm text-slate-600 mt-2">The Relationship Graph is encoded as JSON n
+        <pre class="mt-3 p-3 bg-white rounded text-sm text-slate-800 overflow-auto"><code>
+    augmented = list(candidates)
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+    return augmented</code></pre>
+    </details>
    <button onclick="runSync()" class="gradient-btn px-8 py-4 rounded-xl font-bold flex items-center"
        <i data-lucide="refresh-ccw" class="w-5 h-5"></i> Execute Full Vector Sync
    </button>
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